

Rupak Dey

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EDUCATION

New Mexico State University, Las Cruces, NM

May 2026

Bachelor of Science in Computer Science (Artificial Intelligence) | Minors: Mathematics and Economics

- **GPA:** 3.969/4.00 | **Awards:** Honors, Crimson Scholar, College of A&S Dean's List, International Competitive Scholarship
- **Relevant Courses:** Artificial Intelligence, Data Mining, Deep Learning, Linear Algebra, Statistics, Operating Systems

TECHNICAL SKILLS

Languages: Python, C++, C, Java, SQL, HTML/CSS, JavaScript, TypeScript | **Libraries:** PyTorch, NumPy, Pandas, Scikit-learn, Rasterio, GeoPandas, Matplotlib | **Frameworks & Tools:** Django, Flask, Next.js, MySQL, SQLite, Git/GitHub, Linux

CONFERENCE PRESENTATIONS

[1] L. D. V. Neakrase, N. Chanover, V. L. Hartwick, L. Huber, S. Ajo-Montaño, **R. Dey**, and N. Miller, "The PDS Atmospheric Modeling Annex: A new repository for modeling output," to be presented at the 58th Annual Meeting of the AAS Division for Planetary Sciences, Spokane, WA, USA, Oct. 25–30, 2026.

[2] L. D. V. Neakrase, N. Chanover, V. L. Hartwick, L. Huber, S. Ajo-Montaño, **R. Dey**, and N. Miller, "The PDS Atmospheric Modeling Annex: Adapting to the Community Need for a Planetary Atmospheres Modeling Output Repository," to be presented at the 2026 AGU Annual Meeting, San Francisco, CA, USA, Dec. 7–11, 2026.

RESEARCH & PROFESSIONAL EXPERIENCE

Machine Learning Research Assistant, Forage Utilization Project

September 2025 - Present

KDD Lab, New Mexico State University, Las Cruces, NM | Advisor: *Dr. Huiping Cao*

- Built 10 LSTM models predicting NPP and biomass change from satellite, precipitation, cattle GPS, and land cover data.
- Designed a multi-branch LSTM over 1.3M records, one encoder per source, raising biomass change R^2 from 0.36 to 0.93.
- Ablation established rainfall and land cover drive nearly all skill; cattle GPS added none, validated against MLP and RF.

Research Assistant/Software Engineer, Atmospheres Node, NASA Planetary Data System (PDS)

May 2024 - Present

New Mexico State University, Las Cruces, NM | Supervisor: *Dr. Lynn Neakrase*

- Develop the ELSA web application to enable planetary scientists to generate and archive metadata for large datasets.
- Lead frontend for the Atmospheric Modeling Annex, a new PDS service archiving planetary atmospheric model data.
- Deployed an AI Assistant, Guided Bundle Creation, and 2FA; rebuilt the ELSA UI in Bootstrap 5 for responsive UX.

SELECTED PROJECTS

pantry.ai - 1st Place, Desert Dev Lab Hackathon, NM | Next.js, TypeScript, SQLite, Google Gemini API

April 2026

- Built a PWA with Gemini Vision to identify pantry items, estimate expiry dates, and suggest recipes before items expire.
- Promoted local produce, generated personalized meal plans, and tracked food waste to encourage sustainability.

Airbnb Price Prediction & Feature Analysis | Python, Pandas, Scikit-learn

August 2025 - November 2025

- Optimized a KNN regression model with Decision Tree feature selection and StandardScaler; validated MAE and R^2 .

"Aggie Find-It" Lost & Found Management System | Java, JavaFX, MongoDB

August 2024 - December 2024

- Developed a full-stack MVC with a JavaFX frontend integrated with MongoDB for real-time item reporting and search.

Compiler Construction | C, Flex, Bison, RISC-V Assembly

August 2024 - December 2024

- Built a C-like compiler with Flex/Bison parsing an AST and a code generator handling stack, recursion, and control flow.

LEADERSHIP & SERVICE

Founding President, CS Student Association (CSSA), New Mexico State University, Las Cruces, NM

August 2025 - May 2026

- Launched hackNMSU, NMSU's first hackathon (55 participants and 18 teams); secured \$5K/year in operational funding.
- Built CSSA into NMSU's main CS community, addressing gaps in engagement through contests, panels, and social events.

Student Ambassador, College of Arts and Sciences, New Mexico State University, Las Cruces, NM

August 2023 - May 2026

- Represented the College on the NMSU Strategic Planning Board and advocated for CS at community events.

Vice-President, Google Developer Student Club, New Mexico State University, Las Cruces, NM

September 2024 - May 2025

- Facilitated bi-weekly meetings for 30+ students, focusing on competitive programming and technical skills.

Peer Mentor, TRIO Student Support Services, New Mexico State University, Las Cruces, NM

August 2023 - May 2024

- Mentored 35 STEM students in understanding and solving Java and C problems, obtaining CRLA Level 1 certification.